VIGNESH K

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PROFILE SUMMARY:

Results-driven Software Engineer with 6+ years of experience building scalable data pipelines, ML infrastructure, and enterprise-grade dashboards. Proven expertise in Python, AWS services, and full-stack development with a track record of delivering high-impact solutions that process billions of data points. Skilled in architecting end-to-end systems, leading regional expansions, and driving operational excellence through automation and real-time monitoring solutions.

EXPERIENCE

NOV,2019 – JUL, 2021

JUNIOR ENGINEER, FORD MOTOR PRIVATE LIMITED

1. Worked in Insights and analytics team where we manage day to day jobs
2. Take care of issues that happens when a job scheduled to pull data from source to destination fails and rectify them
3. Take care of the failures of the jobs by analysing the cause of the failure by using tools like Putty, WinSCP etc and rectify them and trigger them accordingly
4. Also worked in Dashboard development project which visualize the historical analysis of jobs and also the real time tracker of the jobs
5. Dashboard is developed using Python Dash plotly library, i played developer role in developing this dashboard.
6. Key technology we used to develop this dashboard are Python, Dash Plotly Library, REST API's, Alteryx, PCF [for hosting] and GITHUB[for code repository], SQLITE3 for DB, Flask-jwt for authentication of API's, flask-restful for structuring the REST API flow
7. I have also played leadership role in mentoring new joinee by providing necessary Knowledge transfer to kick start their stint with the team.

JUL,2021 – Aug, 2022

SOFTWARE ENGINEER II, SPIN GAMES LLC

PROJECT-I – DEVELOPMENT OF CASINO GAMES AND WALLET INTEGRATION

1. Currently working as backend developer in developing casino games like slot, table etc
2. Have profound experience in python backend development especially in handling multiple routes and DB handling wherever required
3. Have done many API integration projects which involves wallet integration with various casino operators
4. Have worked on various game logics involving mathematical inputs
5. Worked on backend technologies python, Flask for routing, Mongo DB for DB related activities, deploying games in EC2 ubuntu machines

PROJECT-II – DEVELOPMENT OF GAME LOBBY USING DJANGO

1. Worked In developing Games lobby portal for internal teams to utilize and test games
2. Project was developed using Django module
3. The portal involves various sections like game, users, install of game versions
4. Developed view sets for above tabs to facilitate front end request using Django libraries like DRF, Djongo [Mongo db connectivity], Token for authentication of API

5. Front end for this module was built using React JS, HTML, CSS

AUG, 2022 – TILL PRESENT

SOFTWARE DEVELOPMENT ENGINEER II, AMAZON WEB SERVICES

Key Projects & Impact:

Bedrock Platform & Infrastructure

Led successful expansion of Amazon Bedrock across 5 new regions (MXP, ZAZ, ARN, HYD, KIX), managing both FES Control Plane and Data Plane components

Resolved critical canary test failures across all regions ensuring service reliability and securing Operational Readiness Review (ORR) sign-offs

Designed and implemented Sherlog Firehose ingestion system capturing billions of request logs from all Bedrock regions into centralized S3 data lake

Created RMS-based subscription filter and CDK infrastructure enabling AWS Fraud team to identify fraudulent activities and build customer insights dashboards

Real-Time Monitoring & Analytics Dashboards

Architected and developed Real-Time Customer Impact Dashboard with React UI, integrating CloudWatch Metrics API across multiple regions/accounts

Built Bedrock Service Availability Dashboard with automated 5-minute data extraction pipeline, programmatic Glue Crawler scheduling, and multi-granularity views (yearly, monthly, daily, 5-minute, custom range)

Created Bedrock Model Availability Dashboard with provider-wise and model-wise breakdowns, enabling targeted analysis of specific model performance

Developed Customer Insights Dashboard with TSDB-based analytics, providing GTM teams visibility into usage patterns, token consumption, and cross-region analytics

Integrated Minerva rate limits (TPM/RPM) into Customer Insights with multi-threaded backend architecture and threshold visualization

Implemented Automated Service Unavailability Alert System with Slack integration, reducing incident response time through real-time 5-minute monitoring windows

Data Annotation & ML Pipeline Enhancement

Built end-to-end Yellowstone annotation tool streamlining vendor data collection for mail threads, creating complete infrastructure, pipeline, CDK, and RDS schema

Implemented inline validation logic increasing throughput and drastically improving data quality by moving validation from offline to online process

Enhanced Bedrock LLM pipeline logging capabilities with S3 logging mechanism across all Spark processing stages, enabling identification of data loss at each stage

Automated RLHF Prompt Collection system scaling from 20-40K annotations/week to 400-600K annotations/day, producing 1M prompts/day

Designed prod-sec approved RLHF Text annotation tool using Annotation Hub and SageMaker Ground Truth

with comprehensive HLD/LLD documentation

Backend Development & Optimization

Developed Online RAG retriever service backend and infrastructure, migrating from Lambda to SageMaker with EC2-hosted model

Reduced latency from 10 seconds to 1-2 seconds, dramatically improving UX with configurable "best of 5" passage retrieval approach

Revamped R+O (RAG + Orchestration) project with JSON editor, validation/fallback logic, and end-to-end Scenario Generation Workflow

Enhanced Horizon Platform with Global CRIS feature, ADC regions support, RAMA insights, latency monitoring, and tier-based UI revamp

Security & Compliance

Implemented comprehensive Prod Sec measures including Supernova domain with Route53 CNAME for API Gateway (TLS 1.2 compliance)

Enabled TLS 1.2 for OpenSearch domain, implemented custom KMS keys for S3 and RDS, and brought backend systems under VPC with VPC endpoints

Technologies: Python, Java, AWS (Lambda, SageMaker, S3, CloudWatch, Glue, RDS, DynamoDB, EC2, CDK), React, REST APIs, Spark, OpenSearch, Docker

EDUCATION

APRIL 2019
BE ECE, RAJALAKSHMI ENGINEERING COLLEGE
Graduated with 7.8 CGPA

APRIL 2015
HIGH SCHOOL, SRI SANKARA VIDYASHRAMAM MHSS
Graduated with 90.8%

SKILLS

Programming Languages: Python, Java, JavaScript, Node JS

Frameworks & Libraries: Django, DRF, Flask, React, Next.js, Dash Plotly

AWS Services: Lambda, SageMaker, S3, CloudWatch, Glue, RDS, DynamoDB, EC2, CDK, Firehose, AppConfig

Databases: MongoDB, SQLite3, RDS, DynamoDB, OpenSearch

Tools & Technologies: Docker, Git, REST APIs, Spark, Kafka, Nifi, SQOOP, Hadoop

Concepts: Data Structures & Algorithms, OOP, Distributed Systems, ML Infrastructure, ETL Pipelines

ACCOMPLISHMENTS

1. Awarded 3rd prize in ITO Innovation ML Hackathon event for the implementation of idea of "Predicting FP failures using Time Series Algorithm"
2. Awarded for hosting ITO Floor event
3. Awarded 2nd prize in coding contest for developing "Attendence Monitoring System using BST

algorithm"
4. Participated in various Transition activities and have profound knowledge on technologies like Kafka, Nifi, SQOOP, Spark, Hadoop
5. Got various appreciation from Senior Managers for developing the dashboard mentioned in Work History section

CERTIFICATIONS

Python, Udemy - 2020
Dash Plotly Visualizations, Udemy - 2020
DS & Algorithms in Python, Udemy – 2021
Flask REST API development, Udemy - 2021

PERSONAL DETAILS

DOB: 17/01/1998
PAN Number: BKWPVR5437R